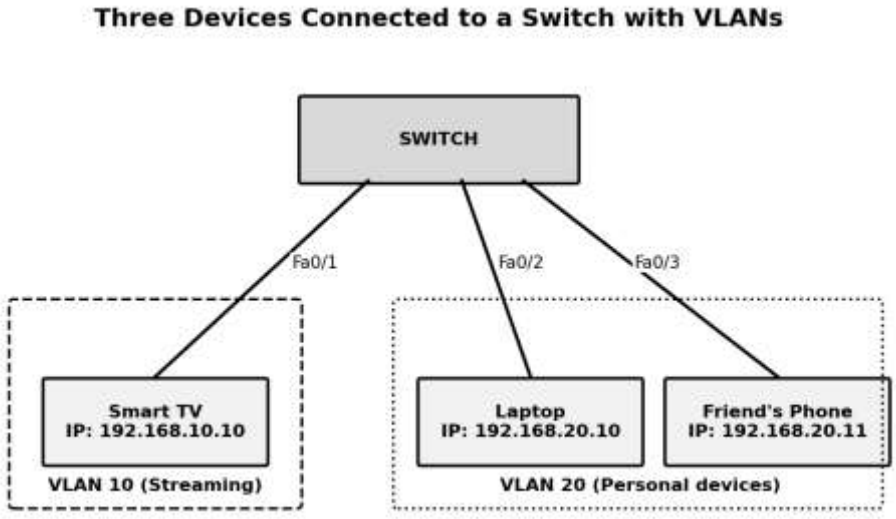
Switching and VLANs

Q1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



Device	IP address	Switch port	VLAN	Reason
Smart TV	192.168.10.10/24	Fa0/1	VLAN 10 (Streaming)	Streaming traffic is kept separate from other devices.
Laptop	192.168.20.10/24	Fa0/2	VLAN 20 (Personal)	Personal and work data stays in its own network.
Friend's phone	192.168.20.11/24	Fa0/3	VLAN 20 (Personal)	Connects through a WiFi access point plugged into this port.

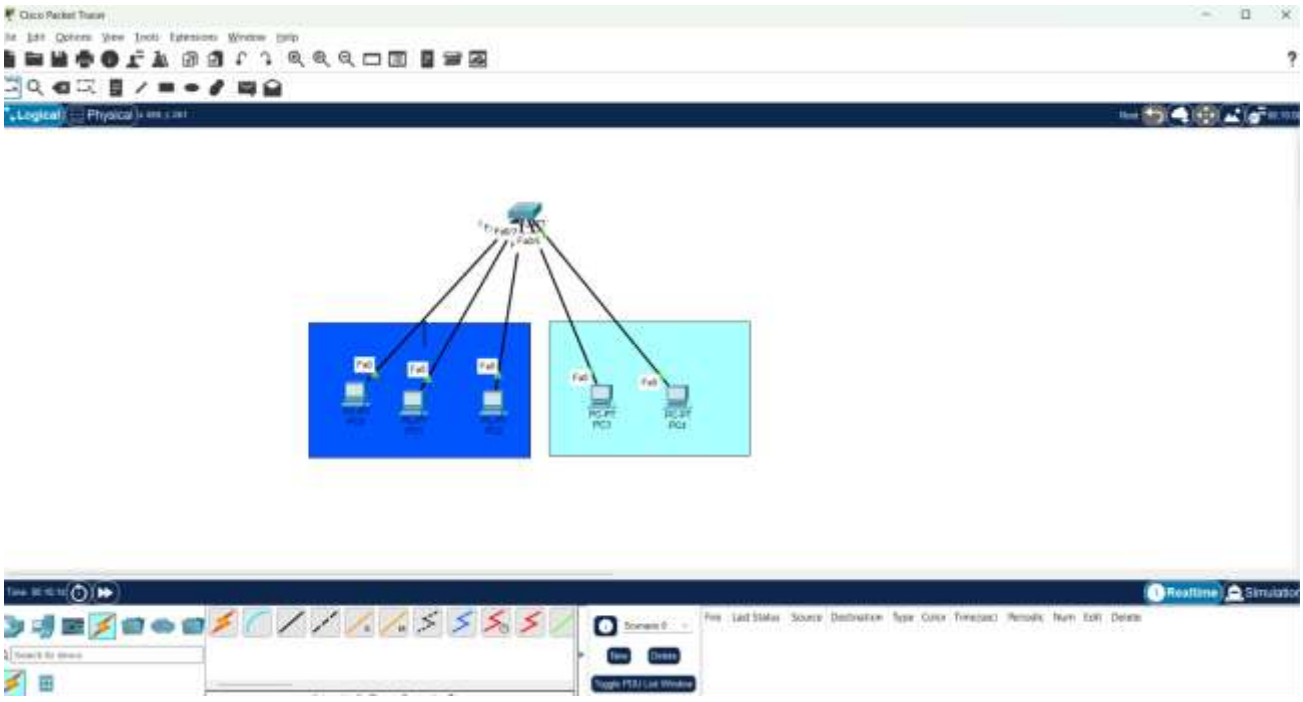
A VLAN (Virtual LAN) divides one physical switch into separate logical networks. Devices in the same VLAN can talk to each other directly, while devices in different VLANs cannot communicate unless a router (or Layer 3 switch) connects them. Here the smart TV cannot reach the laptop or the phone, which improves security and reduces unnecessary broadcast traffic.

Q2. Configure a simulated switch using Cisco Packet Tracer or Net Sim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Use the 'vlan' and 'switchport access vlan commands in the simulator CLI.

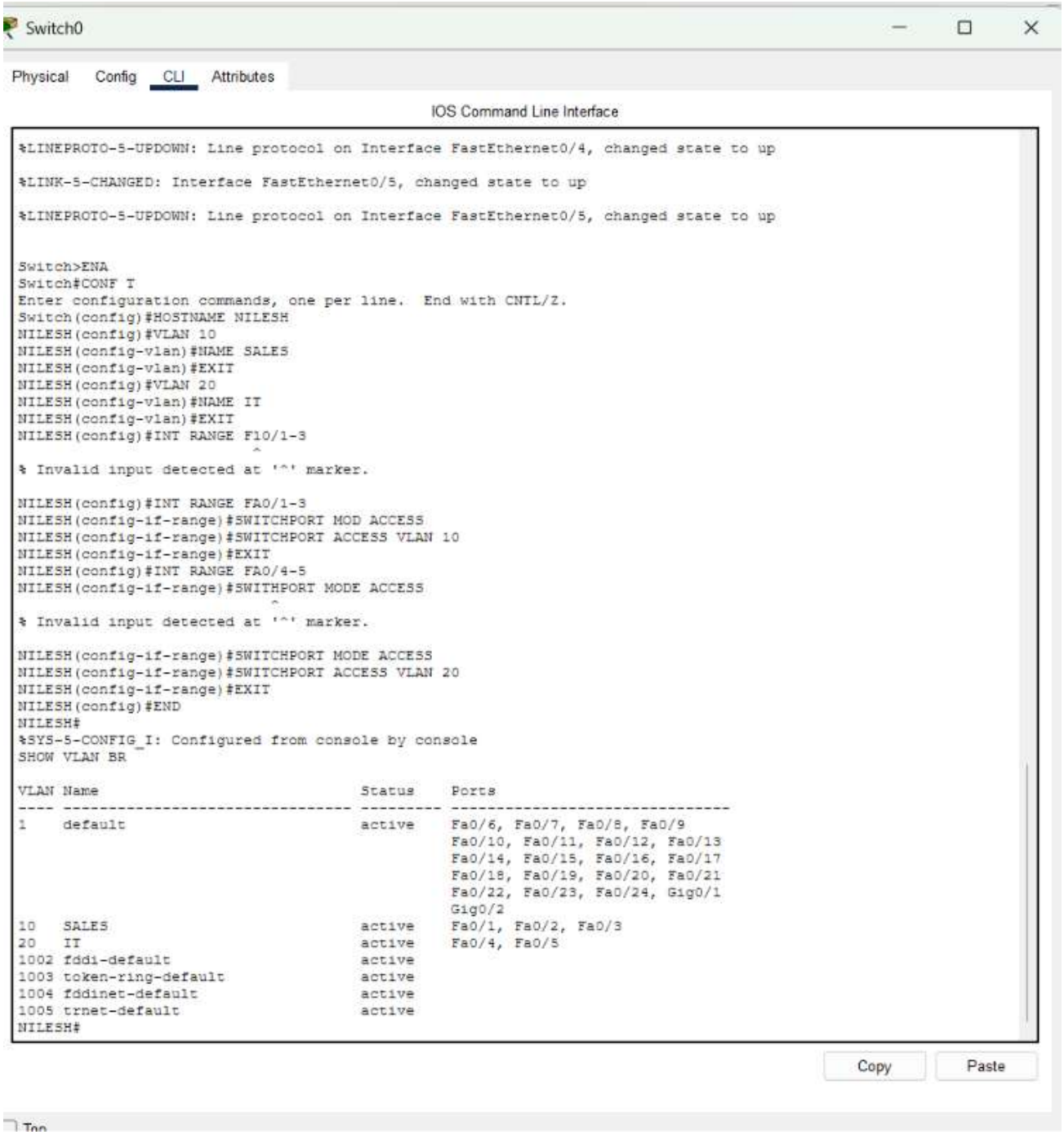
Topology and addressing

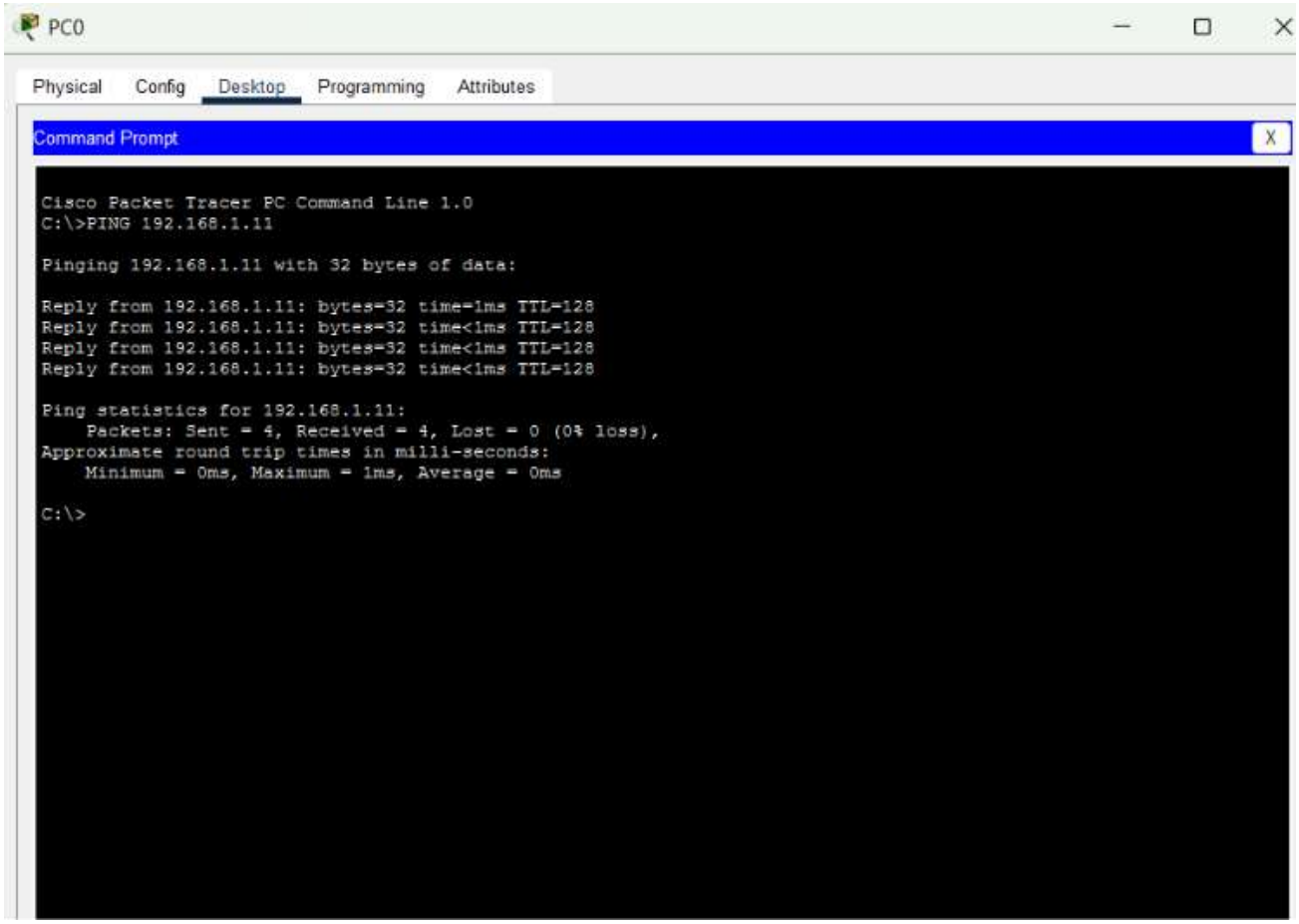
PC	IP address	Subnet mask	Switch port	VLAN
PC1	192.168.1.10	255.255.255.0	Fa0/1	VLAN 10
PC2	192.168.1.11	255.255.255.0	Fa0/2	VLAN 10
PC3	192.168.1.12	255.255.255.0	Fa0/3	VLAN 10

PC	IP address	Subnet mask	Switch port	VLAN
PC4	192.168.2.10	255.255.255.0	Fa0/4	VLAN 20
PC5	192.168.2.11	255.255.255.0	Fa0/5	VLAN 20



Switch configuration commands





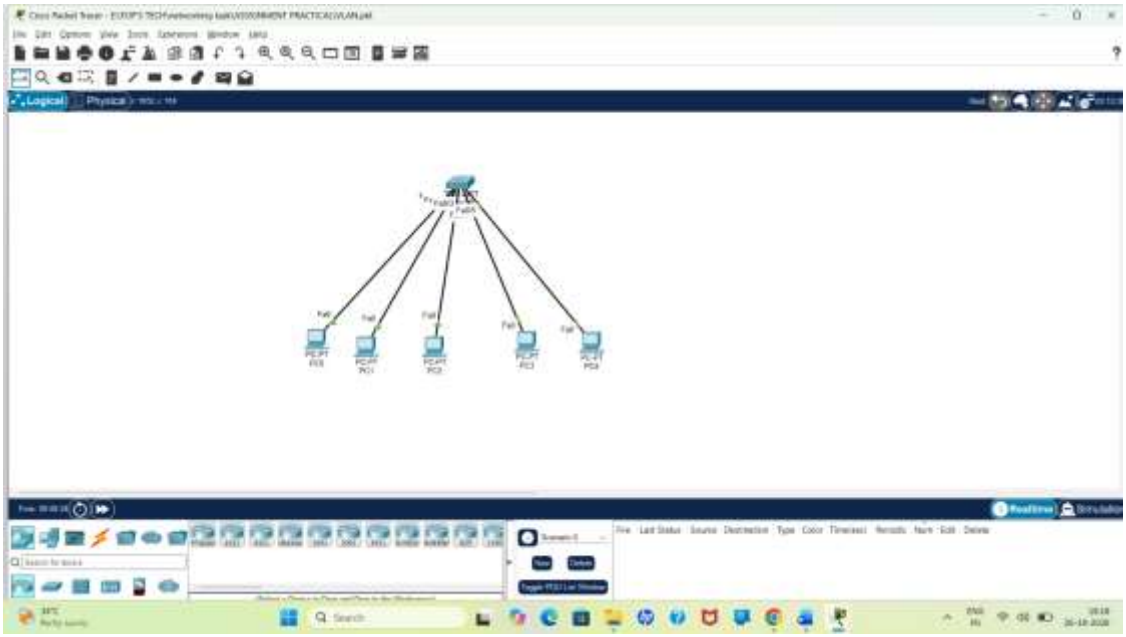
Q3.

In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

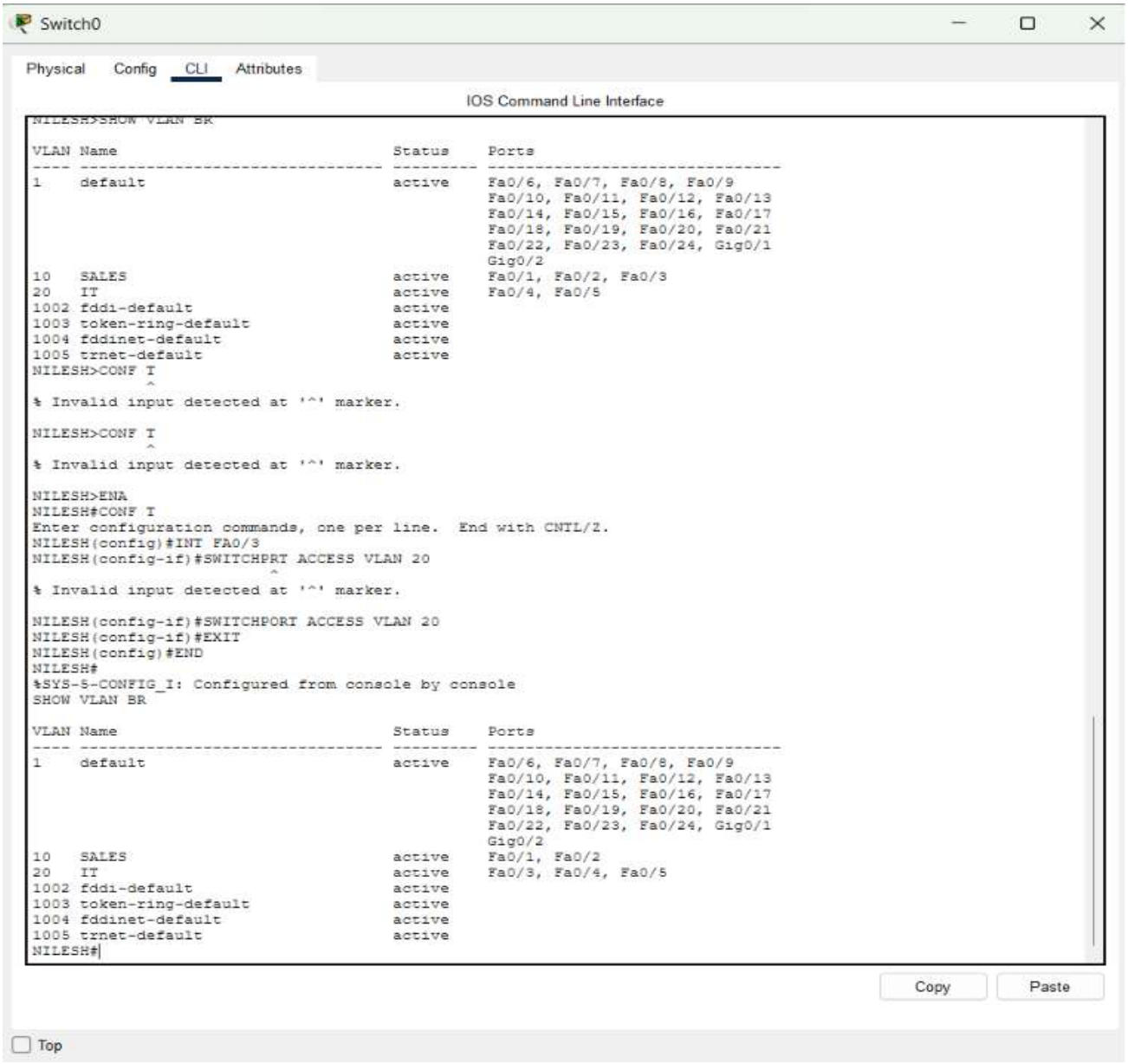
Scenario: PC3 (port Fa0/3) is moved from VLAN 10 to VLAN 20.

Before the change

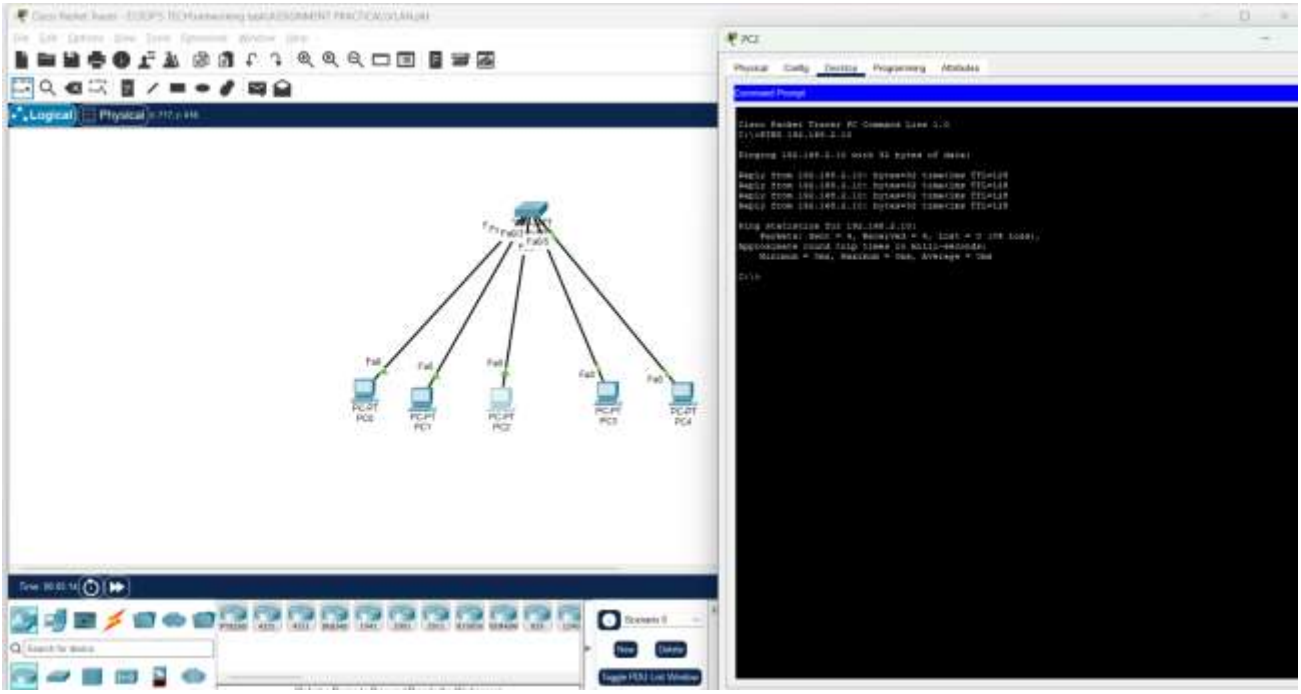
```
SW1# show vlan brief
10   Streaming  active   Fa0/1, Fa0/2, Fa0/3
20   Personal   active   Fa0/4, Fa0/5
```



Change commands



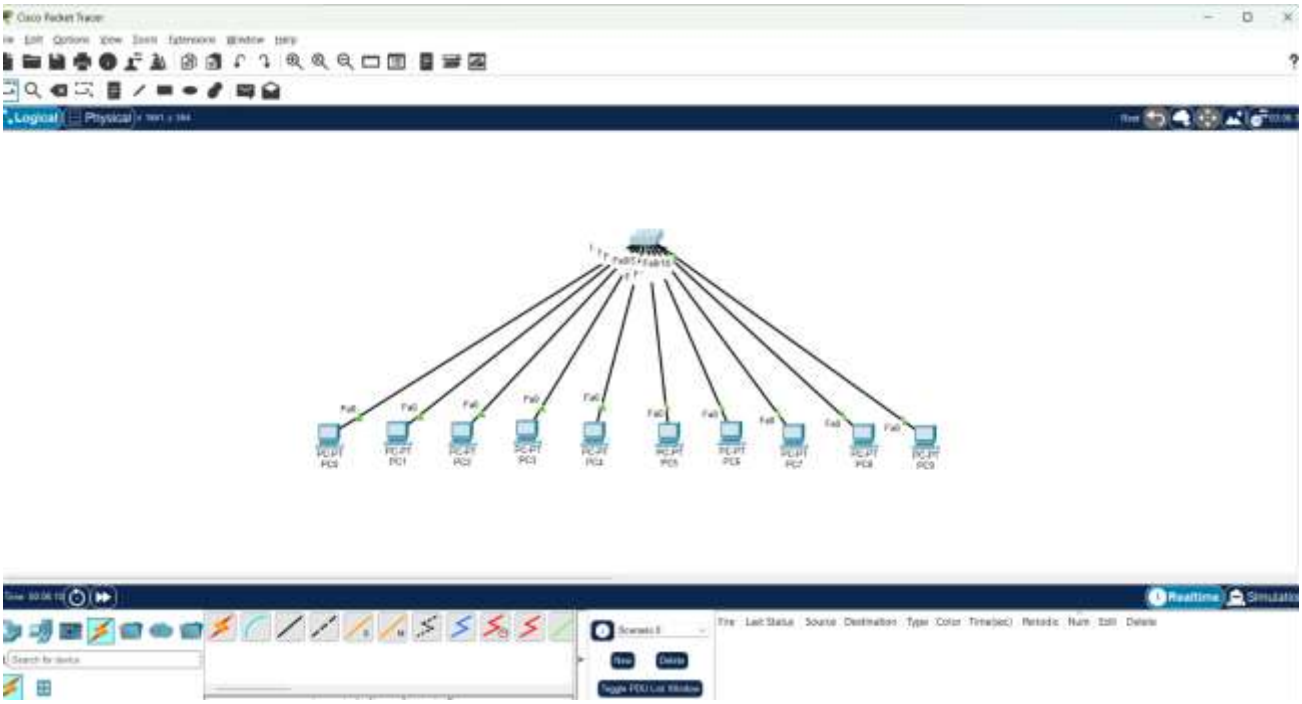
From	To (ping)	Expected result	Reason
PC3 (now VLAN 20)	PC4 192.168.20.1	Success	Now in the same VLAN and subnet
PC3 (now VLAN 20)	PC5 192.168.20.2	Success	Same VLAN and subnet
PC3 (now VLAN 20)	PC1 192.168.10.1	Fails	Different VLANs
PC1 (VLAN 10)	PC3 192.168.20.3	Fails	Different VLANs

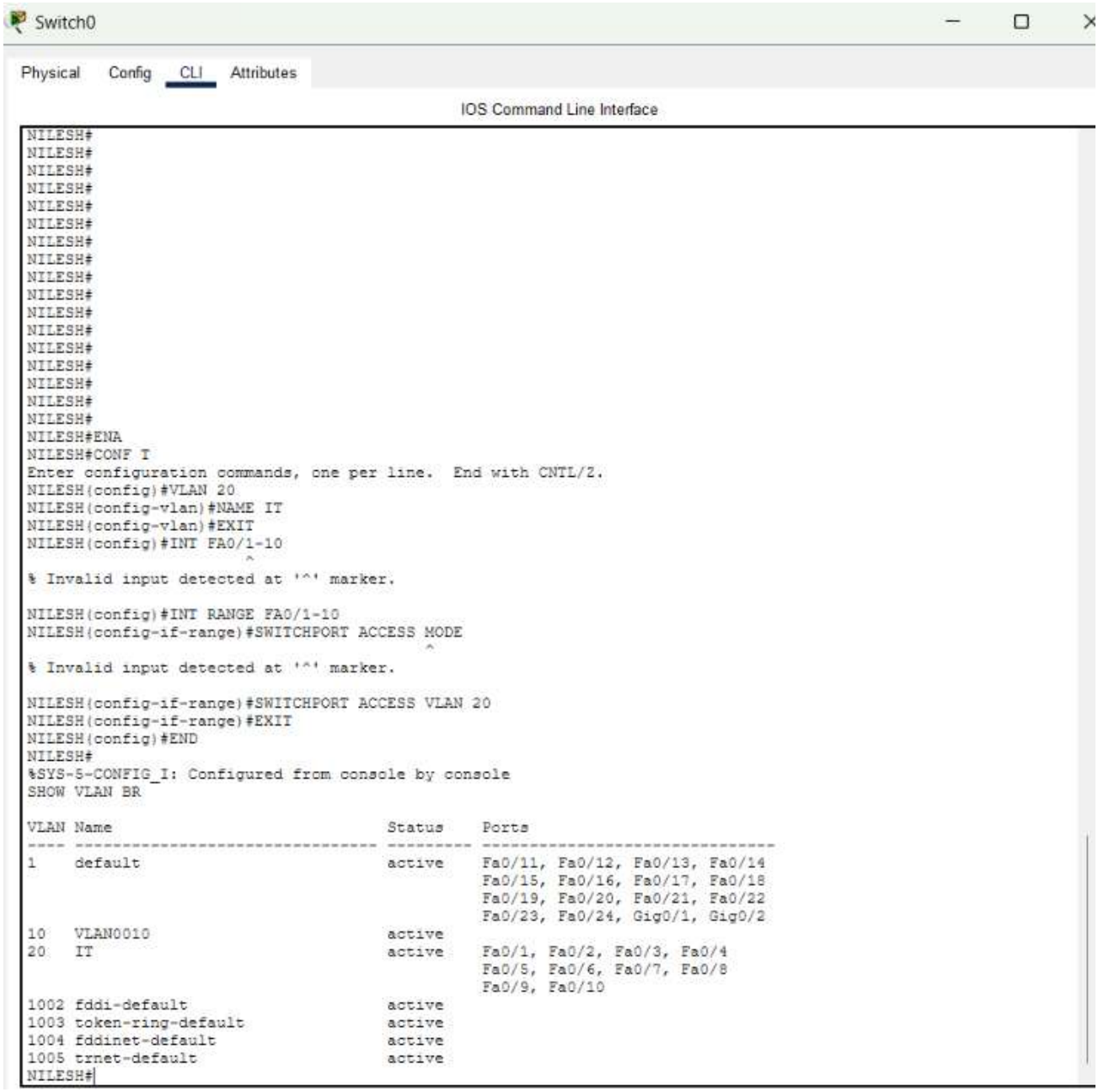


Q4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3). Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

Simulator: Cisco Packet Tracer (the same commands also work in GNS3 with Cisco IOS images).

Method: The interface range command applies the same commands to ports Fa0/1 to Fa0/10 together, so the commands are written only once instead of 10 times.





Explanation

- vlan 30 / name Guests: creates VLAN 30 (if the VLAN does not exist, Packet Tracer also creates it automatically when it is assigned to a port).
- interface range fastEthernet 0/1 - 10: selects the 10 ports Fa0/1 to Fa0/10 at once. (Another style that works is: interface range fa0/1 - 10.)
- switchport mode access and switchport access vlan 30: set the 10 ports as access ports of VLAN 30.
- no shutdown: makes sure that the ports are turned on.
- show vlan id 30: checks that Fa0/1 to Fa0/10 are listed in VLAN 30.